Distributed Order Management

Classical and Quantum Optimization for Divert Recommendations

Nestle WISER Quantum + AI Program 2026

The Problem

- A focus order is one the default distribution center cannot fully satisfy - inventory runs out, dock capacity is full, or a rule blocks it.
- Diverting to another DC costs more to ship, and only helps if the improvement clears a threshold.
- Every focus order competes with others for the same shared inventory and dock slots.

409

of 1,109 orders are focus orders - their default DC cannot fully cover demand

14

candidate distribution centers to choose from per divert

Approach

Classical: exact MILP

PuLP / CBC, proves optimality. 7 constraints: one DC per order, demand limits, pooled inventory with a 5-day lookahead, divert threshold, case/pallet decomposition, dock capacity, penalty logic.

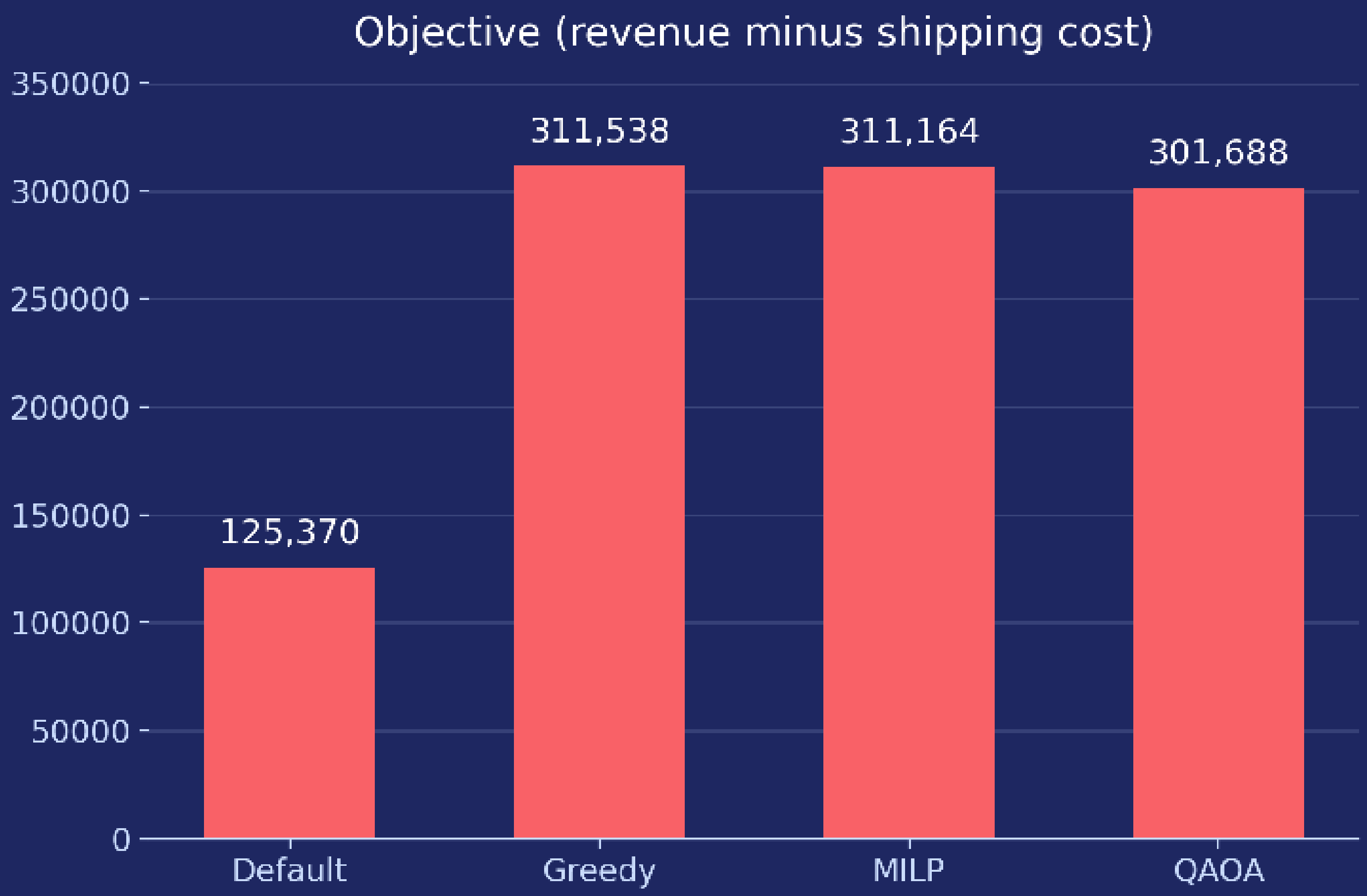
Quantum: QAOA

Qiskit implementation, QUBO encoding of the one-DC-per-order constraint. Multi-restart COBYLA against an exact statevector simulation. Capped at 15 qubits - simulation cost doubles per qubit.

Baselines

Default (no diverts) and an order-level greedy heuristic - the floor both solvers need to clear.

Results: 5 Focus Orders x 3 DCs



0.80 fill rate

MILP - highest of any strategy tested

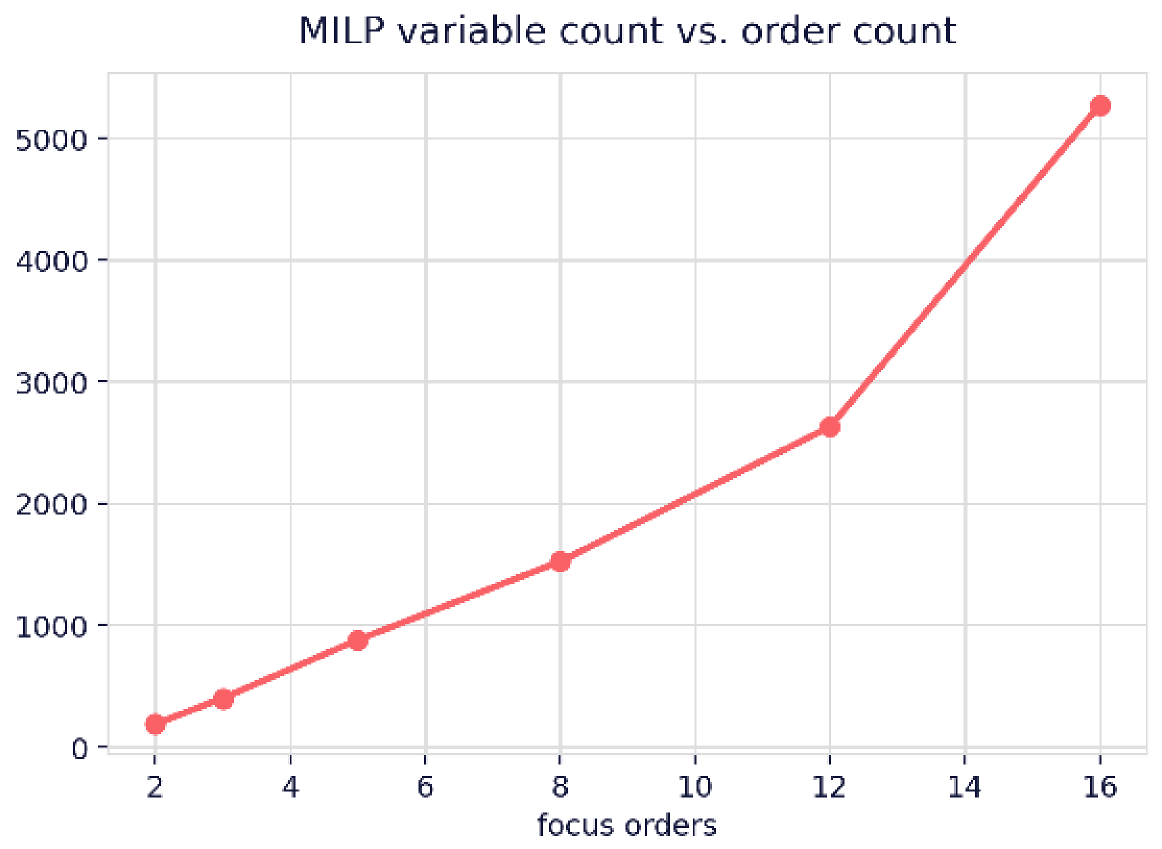
0 of 5

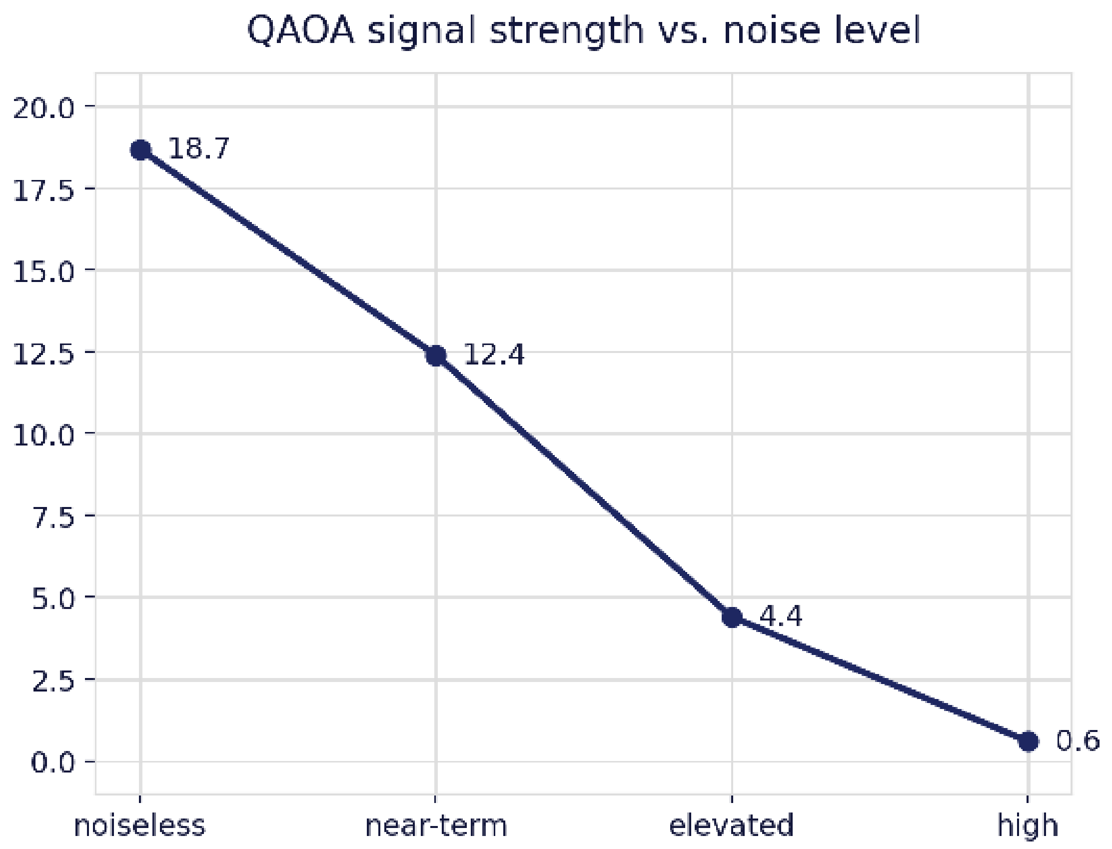
QAOA configs pass the full feasibility check

0.34 -> 0.80

fill rate, default vs. MILP recommendation

Scaling and Noise





Limitations and Next Steps

Limitations

- QAOA and greedy objectives are not checked against the full constraint set; QAOA fails every tested feasibility check
- Case-pick/pallet-pick capacity ceilings are not enforced - usage history exists, a stated limit does not
- MILP is untested at full order volume (~300+ focus orders/day)
- QAOA results are simulator-only

Next Steps

- Encode QAOA's missing constraints as QUBO penalty terms - qubit cost already quantified
- Real capacity data for case-pick/pallet-pick limits
- Commercial solver or region/SKU-family batching for full order volume
- A hardware run, at reduced circuit depth, once the above is in place